

Reparametrization Invariant Amplitudes in Heavy Particle Effective Theories

Zhi-Zhong Li,^c Ian Low,^{b,c} Ming-Lei Xiao^{a,b,c}

^a*School of Science, Sun Yat-sen University,
Shenzhen 518100, China*

^b*High Energy Physics Division, Argonne National Laboratory,
Argonne, IL 60439, U.S.A.*

^c*Department of Physics and Astronomy, Northwestern University,
Evanston, IL 60208, U.S.A.*

E-mail: Zhi-zhongLi2024@u.northwestern.edu, ilow@northwestern.edu,
xiaomlei@mail.sysu.edu.cn

ABSTRACT: We investigate the reparametrization invariance (RPI) of Heavy Quark Effective Theory (HQET) through the lens of scattering amplitudes. By constructing a basis of heavy quark amplitudes with gluons, we systematically impose RPI constraints, gauge invariance, and charge conjugation symmetry to identify the independent physical degrees of freedom up to $O(m^{-3})$. A direct mapping between these amplitudes and the HQET Lagrangian is established, demonstrating consistency with known operator bases. Our approach bridges on-shell amplitude methods with effective field theory techniques, offering a streamlined framework for studying heavy quark dynamics and extending naturally to other non-relativistic systems, such as dark matter or beyond-the-Standard-Model scenarios.

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1 Introduction

Recent years have seen developments in constructing the effective field theories for massless degrees of freedom, such as the Standard Model Effective Field Theory (SMEFT) in the unbroken phase of the electroweak symmetry [1–4] and the chiral effective theories for Goldstone bosons [5–7] in either Quantum Chromodynamics (QCD) or Beyond the Standard Model (BSM) theories [8, 9]. Massive particles are allowed in a limited cases, like the fermions in the Low-energy Effective Field Theory [10, 11] (LEFT, also known as the Weak Effective Theory, WET [12]) which are still treated in a Lorentz invariant fashion. In all these cases, an approach that proves to be efficient is to construct the independent set of “building blocks” of on-shell amplitudes that are allowed by the symmetries of the model, which turns out to be isomorphic to the operator basis of the EFT [1, 7, 13–15]. The most obvious advantage is that the Equation of Motion (EOM) redundancy is absent in the construction due to the on-shell condition of the external particles. Furthermore, when written in terms of spinor-helicity variables, the independent set of on-shell amplitudes can be enumerated systematically using the Young Tensor method [1, 15, 16], which resolves all the redundancy relations including the momentum conservation (or the integration by part for the operators) and the Lorentz Schouten Identities.

However, a Lorentz non-invariant formalism is convenient in scenarios involving the non-relativistic interactions between particles with hierarchical scales, such as the decay of a $Q\bar{q}$ meson with $m_Q \gg \Lambda_{\text{QCD}}$ and $m_q \lesssim \Lambda_{\text{QCD}}$. The momentum exchange is typically of the scale Λ_{QCD} which is significantly smaller than m_Q . To avoid large $\log(m_Q/\Lambda_{\text{QCD}})$ corrections, the Heavy Quark Effective Theory (HQET) is introduced, characterized by its $1/m_Q$ expansion.

In a frame independent manner, a reference 4-velocity v^μ with $v^2 = 1$ is defined in the heavy quark momentum $p^\mu = m_Q v^\mu + k^\mu$, so that the deviation k^μ is perturbative relative to the mass m_Q . The effective Lagrangian is thus written according to the power counting of $1/m_Q$ that depends on v^μ , and so are the effective operators [17–19]. Such feature is shared by a variety of effective field theories, including the EFT approach to Dark Matter (DM) model building [20–22] and Black Hole (BH) physics [23, 24]. For example, when the DM is assumed to be heavier than SM particles, a natural EFT construction of the DM-SM interactions would be very similar to the HQET, where a velocity vector v^μ of the DM particle is chosen to break the Lorentz invariance. In this paper, all the EFTs with such feature are called Heavy Particle Effective Theory (HPET) [17, 24].

In the HPET formalism, the reference velocity v^μ (which is often taken to be time direction) spontaneously breaks the Lorentz symmetry, rendering the effective operators non-invariant under Lorentz transformations. Consequently, the corresponding on-shell amplitudes also lose their Lorentz invariance, which precludes the direct application of the Young Tensor method. Therefore, new techniques are required for an on-shell construction of such effective field theories.

The problem is very similar to that of chiral perturbation theory (ChPT), where a global symmetry is spontaneously broken and realized non-linearly in the EFT. In ChPT, we impose the Adler Zero condition on the on-shell amplitudes to get an independent set of “soft blocks” that correspond to the higher derivative corrections in the Lagrangian [6, 7]. As a result, cross-dimensional constraints among effective operators are reproduced in the corresponding amplitudes via the Adler Zero condition. The idea is similar in HPET: we need to impose some conditions on the on-shell amplitudes constructed from the low-energy degrees of freedom, that should reflect the non-linearly realized Lorentz symmetry; and this should be the reparameterization invariance (RPI), which says that observables derived from $\mathcal{L}_v$ should be independent of the choice v . In other words, under the reparameterization $v^\mu \rightarrow w^\mu = v^\mu + \ell^\mu/m$ where $w^2 = 1$, on-shell amplitudes should be invariant.

In this paper, we are going to construct such an RPI amplitude basis and reproduce the corresponding well-known operator basis of HQET up to the order $1/m^3$. In section 2, we build the heavy quark amplitudes with gluons, both on-shell and off-shell, purely from bottom up. In section 3, we perform the RPI computations and find the allowed combinations of the amplitudes; we also demonstrate how they can be inferred from a top-down approach. In section 4, we provide the dictionary between the amplitudes and the effective operators, and show that the result indeed corresponds to the HQET Lagrangian. We conclude and comment on the application of our method at the end. We also include an appendix to briefly introduce the HQET Lagrangian.

2 Amplitudes of Heavy Quarks with a Gluon

2.1 Building blocks of heavy quark amplitudes

In general, a scattering amplitude depends on the wave functions of the external particles and their kinematic variables. It is sometimes convenient to introduce the spinor variables $(\lambda_i, \tilde{\lambda}_i)$ to describe both of them when the particles are on-shell, either massless or massive [25]. However, when the anti-particle is integrated out, spinor variables may not be the most convenient choice¹. In particular, the space of massive Dirac spinors is decomposed into subspaces of particles and anti-particles according to the projection operators

$$P_{\pm} = \frac{1}{2}(1 \pm \not{v}) , \quad (2.1)$$

where v_{μ} is a time-like unit vector $v^2 = 1$ indicating a reference “time direction”, typically chosen to be close to the 4-velocity of the heavy particle p_{μ}/m . Therefore, a heavy fermion is described by a heavy spinor variable u_v constrained in the P_+ subspace, and thus satisfies $P_- u_v = 0$, or

$$\not{v} u_v = u_v . \quad (2.2)$$

On the other hand, the momentum is decomposed as $p_{\mu} = mv_{\mu} + k_{\mu}$ where k_{μ} is the low energy kinematic variable, similar to the 3-momentum in the rest frame. The energy-momentum relation indicates

$$m^2 = p^2 = (mv + k)^2 = m^2 + 2mv \cdot k + k^2 , \quad \Rightarrow \quad v \cdot k = -\frac{k^2}{2m} . \quad (2.3)$$

Eq. (2.2) and (2.3) together constitute the on-shell condition of a heavy fermion.

When we use the heavy spinor $u_v(k)$ to construct local amplitudes, eq. (2.2) puts extra constraints. In particular, all the fermion bilinear should be understood as a projection onto the P_+ subspace $P_+ \Gamma P_+$, and the Clifford algebra is reduced to the basis

$$\begin{aligned} P_+ \Gamma P_+ &= a P_+ + b_{\mu\nu} P_+ (-i\sigma^{\mu\nu}) P_+ , \\ a &= \frac{1}{2} \text{tr}(P_+ \Gamma) , \quad b_{\mu\nu} = -\frac{i}{4} \epsilon_{\mu\nu\rho\lambda} v^{\rho} \text{tr}(\Gamma P_+ \gamma^{\lambda} \gamma^5 P_+) . \end{aligned} \quad (2.4)$$

Therefore, it is convenient to define the following two building blocks

$$S_v \equiv \bar{u}_v u_v , \quad T_v^{\mu\nu} \equiv \bar{u}_v (-i\sigma^{\mu\nu}) u_v = -i\epsilon^{\rho\mu\nu\lambda} v_{\rho} (\bar{u}_v \gamma_{\lambda} \gamma^5 u_v) . \quad (2.5)$$

where $T_v^{\mu\nu}$ is anti-symmetric and transverse $T_v^{\nu\mu} = -T_v^{\mu\nu}$, $T_v^{\mu\nu} v_{\nu} = 0$.

Another important feature of the heavy spinors is their normalization. Eq. (2.2) only determines u_v up to a scalar factor, and unlike the relativistic case where the only scalars

¹A description based on modified heavy spinor variables is introduced in [26]

available are constants, this factor could depend on the kinematic variable $f(k^2)$ ². In general, we have a solution

$$u_v(k) = f(k^2)P_+u(p) \quad (2.6)$$

where $u(p)$ is the Lorentz covariant Dirac spinor with momentum $p = mv + k$. Note that the Dirac equation $(\not{p} - m)u = 0$ indicates $P_-u = \frac{\not{k}}{2m}u$, we obtain

$$\begin{aligned} u_v(k) &= f(k^2)(1 - P_-)u(p) = f(k^2) \left(1 - \frac{\not{k}}{2m}\right) u(p) \\ u(mv + k) &= \left[f(k^2) \left(1 - \frac{k^2}{4m^2}\right) \right]^{-1} \left(1 + \frac{\not{k}}{2m}\right) u_v(k) \end{aligned} \quad (2.7)$$

It is often convenient to choose $f(k^2) = 1$, while other choices may also be interested. For example, consider the Lorentz transformation $w_\mu = \Lambda(w, v)_\mu^\nu v_\nu$, which has its spinor representation as []

$$\tilde{\Lambda}(w, v) = \frac{1 + \not{w}\not{v}}{\sqrt{2(1 + w \cdot v)}}. \quad (2.8)$$

If we take $w_\mu = v_\mu + k_\mu/m = p_\mu/m$, we have

$$\tilde{\Lambda}(v + k/m, v) = \frac{2 + \not{k}\not{v}/m}{2\sqrt{1 + \frac{v \cdot k}{2m}}} = \left(1 - \frac{k^2}{4m^2}\right)^{-1/2} \left(1 + \frac{\not{k}\not{v}}{2m}\right). \quad (2.9)$$

Hence

$$\begin{aligned} \tilde{\Lambda}(v + k/m, v)u_v(k) &= f(k^2) \left(1 - \frac{k^2}{4m^2}\right)^{-1/2} \left(1 + \frac{\not{k}}{2m}\right) \left(1 - \frac{\not{k}}{2m}\right) u(p) \\ &= f(k^2) \left(1 - \frac{k^2}{4m^2}\right)^{1/2} u(p), \end{aligned} \quad (2.10)$$

which is a k -dependent scaling of the original spinor u . Thus we may also take $f(k^2) = (1 - k^2/4m^2)^{-1/2}$ so that $u_v(k)$ is simply a k -dependent Lorentz transformation of $u(p)$. Different choices of the normalization would lead to the different values of the building blocks in eq. (2.5), and they reparameterize differently as we will show later. Such redundancy of normalization corresponds to the freedom of field redefinition in the Lagrangian formalism.

Regarding power counting of the HPET, we conventionally use the $|k_\perp| \ll m$ where $k_\perp \equiv k - v(v \cdot k)$. Eq. (2.3) indicates that the temporal component $v \cdot k$ is at higher order than $k_\perp$. Therefore we define the perturbative parameter as

$$\alpha_i \equiv \frac{1}{4m^2}k_{\perp i}^2 = \frac{1}{4m^2}[k_i^2 - (v \cdot k_i)^2], \quad (2.11)$$

²Another scalar $v \cdot k$ is related to k^2 by eq. (2.3), while $v^2 = 1$ by definition.

such that

$$\begin{aligned} k_i^2 &= 4m^2\alpha_i + 4m^2\alpha_i^2 + O(\alpha_i^3) , \quad v \cdot k_i = -2m\alpha_i - 2m\alpha_i^2 + O(\alpha_i^3) , \\ k_1 \cdot k_2 &= \frac{1}{2}[k_1^2 + k_2^2 - (k_1 - k_2)^2] = 2m^2(\alpha_1 + \alpha_2) - \frac{1}{2}q^2 + O(\alpha_i^2) . \end{aligned} \quad (2.12)$$

We have adopted the shorthand notation

$$k = \frac{1}{2}(k_1 + k_2) , \quad q = k_1 - k_2 \quad (2.13)$$

where k is the average momentum shift of the heavy particle and q is the momentum transfer. In sum, a general amplitude involving a single heavy fermion could be written as

$$\mathcal{M}_v = AS_v + B_{\mu\nu}T_v^{\mu\nu} \quad (2.14)$$

where the form factors A, B may depend on v , the momenta $k_{1,2}$, and the wave functions of any other particles in the interaction, while they series expand in powers of $\alpha_{1,2}$.

2.2 Bottom-up approach and reparametrization invariance

The effective field theory constitutes of all possible effective operators under the symmetry constraints. From the on-shell point of view, it amounts to all possible local amplitudes. For heavy particle amplitudes in terms of the heavy spinors $u_v(k)$, the Lorentz symmetry is spontaneously broken by the choice of v^μ . Similar to the case of non-linear sigma model, such spontaneously broken symmetry imposes non-linear constraints on the effective operators or the local amplitudes. The constraint on the HPET can be described by the reparameterization invariance (RPI), which says that physical observables do not depend on the artificial choice of v^μ . Suppose we chose another reference vector $w^\mu = v^\mu + \ell^\mu/m$, such that the momentum variables are redefined accordingly

$$k'^\mu_i = k^\mu_i - \ell^\mu , \quad k'^\mu = k^\mu - \ell^\mu , \quad q'^\mu = q^\mu . \quad (2.15)$$

The heavy spinor changes according to eq. (2.7)

$$\begin{aligned} u_w(k') &= f(k'^2) \left(1 - \frac{\not{k}'}{2m}\right) u(p) = \frac{f(k'^2)}{f(k^2)} \frac{1}{1 - \frac{k^2}{4m^2}} \left(1 - \frac{\not{k}'}{2m}\right) \left(1 + \frac{\not{k}}{2m}\right) u_v(k) \\ &= \frac{f(k'^2)}{f(k^2)} \left[1 + \frac{1}{1 - \alpha} \frac{\not{\ell}}{2m} \left(1 + \frac{\not{k}}{2m}\right)\right] u_v(k) + O(\alpha^2). \end{aligned} \quad (2.16)$$

In this section, we will take the normalization $f(k^2) = 1$ for convenience. Using the above relations, the new amplitude for the reference vector w^μ can be written as a function of the original variables and the shift parameter ℓ^μ

$$\mathcal{M}(u_w, w, k') \equiv \widetilde{\mathcal{M}}(u_v, v, k; \ell) \quad (2.17)$$

RPI then requires that the amplitude be independent of the shift

$$\mathcal{M}(u_v, v, k) = \mathcal{M}(u_w, w, k') \quad \Leftrightarrow \quad \widetilde{\mathcal{M}}(u_v, v, k; 0) = \widetilde{\mathcal{M}}(u_v, v, k; \ell) \quad (2.18)$$

Naively one could write down $\partial \widetilde{\mathcal{M}} / \partial \ell = 0$, but note that instead of being arbitrary, ℓ is constrained by the on-shellness of the reference vector

$$w^2 = (v + \ell/m)^2 = 1 \quad \Rightarrow \quad v \cdot \ell = -\frac{\ell^2}{2m} . \quad (2.19)$$

The amplitude only needs to stay constant along the hyperbola $w^2 = 1$ rather than for any shift ℓ^μ . It translates to the actual condition of RPI as

$$\mathcal{D}_\perp^\mu \mathcal{M}_v \equiv (\eta^{\mu\nu} - v^\mu v^\nu) \frac{\partial}{\partial \ell^\nu} \widetilde{\mathcal{M}} \Big|_{\ell=0} = \eta_\perp^{\mu\nu} \frac{\partial}{\partial \ell^\nu} \widetilde{\mathcal{M}} \Big|_{\ell=0} , \quad (2.20)$$

where $\eta_\perp^{\mu\nu} \equiv \eta^{\mu\nu} - v^\mu v^\nu$ is the vector projector onto the transverse subspace. In general, the amplitude $\mathcal{M}_v$ depends on a complete set of effective operators and their Wilson coefficients, while the above condition potentially sets constraints on these parameters. We will be applying the condition to the heavy particle amplitudes to extract independent parameters of the HPET.

2.2.1 Heavy Quark Amplitudes with an on-shell gluon

Now we consider the coupling between the heavy quark and a gauge boson, which can be described by the 3-point on-shell amplitude

$$\mathcal{M}_v(k_1, k_2, \epsilon) = A(v, k, q, \epsilon) S_v + B_{\mu\nu}(v, k, q, \epsilon) T_v^{\mu\nu} , \quad (2.21)$$

where both A, B are functions of the heavy particle momenta $k_{1,2}$ and linearly depend on the polarization vector ϵ . Note that the dot products among v, k_1, k_2 can be cast into polynomials of α_i by eq. (2.12), while the polarization vector satisfies its own on-shell condition $q \cdot \epsilon = 0$, the only two parity-even scalar factors in A, B are $v \cdot \epsilon$ and $k_i \cdot \epsilon = k \cdot \epsilon$ up to a polynomial of the perturbative parameters α_i . The anti-symmetric and transverse tensor $T^{\mu\nu}$ can only contract with k, q, ϵ . Therefore, the coefficient functions A, B have the general form³:

$$\begin{aligned} A &= C_1(\alpha_i)(v \cdot \epsilon) + \frac{C_2(\alpha_i)}{m}(k_\perp \cdot \epsilon) , \\ B_{\mu\nu} &= \frac{1}{8m^2} \left[C_3(\alpha_i)(v \cdot \epsilon) + \frac{C_4(\alpha_i)}{m}(k_\perp \cdot \epsilon) \right] k_\mu q_\nu + \frac{1}{2m} [C_5(\alpha_i)q_\mu + C_6(\alpha_i)k_\mu] \epsilon_\nu . \end{aligned} \quad (2.22)$$

where $C_a(\alpha_i)$ are polynomial functions of α_i . Note that we have chosen $k_\perp \cdot \epsilon \equiv \eta_\perp^{\mu\nu} k_\mu \epsilon_\nu$ in the basis due to the power counting convention, which will prove to be handy when compared

³There are a few parity-odd form factors with Levi-Civita tensor $\epsilon_{\mu\nu\rho\lambda}$ that are not listed.

to the Lagrangian. Therefore, the generic amplitude could be written as a combination of a basis of 6 on-shell amplitudes

$$\mathcal{M}_v(k_1, k_2, \epsilon) = \sum_{a=1}^6 C_a(\alpha_i) M_a , \quad (2.23)$$

which are listed as following

$$\begin{aligned} M_1 &= (v \cdot \epsilon) S_v \\ M_2 &= \frac{1}{m} (k_\perp \cdot \epsilon) S_v \\ M_3 &= \frac{1}{4m^2} (v \cdot \epsilon) k_\mu q_\nu T_v^{\mu\nu} \\ M_4 &= \frac{1}{4m^3} (k_\perp \cdot \epsilon) k_\mu q_\nu T_v^{\mu\nu} \\ M_5 &= \frac{1}{2m} q_\mu \epsilon_\nu T_v^{\mu\nu} \\ M_6 &= \frac{1}{2m} k_\mu \epsilon_\nu T_v^{\mu\nu} . \end{aligned} \quad (2.24)$$

Before imposing RPI, there are a few constraints we want to impose. As we know, QCD also has the discrete symmetry of charge conjugation $\mathcal{C}$, which exchanges particles and anti-particles. In the all-outgoing convention where particle 1 is anti-particle with momentum $-p_1$, it would swap with the particle 2 under the charge conjugation, that amounts to the exchange of the momenta $\mathcal{C}p_1 = -p_2$. With $p_i = mv + k_i$, we have

$$\begin{aligned} \mathcal{C}k_1 &= -k_2 , \quad \mathcal{C}k_2 = -k_1 , \quad \mathcal{C}v = -v , \\ \Rightarrow \quad \mathcal{C}k &= -k , \quad \mathcal{C}q = q , \quad \mathcal{C}\alpha_{1,2} = \alpha_{2,1} . \end{aligned} \quad (2.25)$$

For the wave functions, S_v and T_v are symmetric and anti-symmetric with respect to the swap (including the symmetries of the Lorentz structure and the Grassmann nature of the fermions), while the gauge boson should be odd under charge conjugation, thus

$$\mathcal{C}S_v = S_v , \quad \mathcal{C}T_v = -T_v , \quad \mathcal{C}\epsilon = -\epsilon . \quad (2.26)$$

Together, we can derive how the amplitude change under $\mathcal{C}$. In our case, where we only consider $\mathcal{C}$ -even interactions, we find that $M_{1,2,3,4,5}$ are all $\mathcal{C}$ -even while M_6 is $\mathcal{C}$ -odd. It means that their coefficients should have the same $\mathcal{C}$ parity. For example at $O(\alpha^1)$ order, the $\mathcal{C}$ -even/odd dependence are $\alpha_1 \pm \alpha_2$. Taking the $\mathcal{C}$ symmetry into account, the general local amplitude up to $O(m^{-3})$ would have the form

$$\begin{aligned} \mathcal{M}_v(k_1, k_2, \epsilon) &= [c_1 + c_1^1(\alpha_1 + \alpha_2)] M_1 + [c_2 + c_2^1(\alpha_1 + \alpha_2)] M_2 \\ &\quad + c_3 M_3 + c_4 M_4 + [c_5 + c_5^1(\alpha_1 + \alpha_2)] M_5 + c_6^1(\alpha_1 - \alpha_2) M_6 + O(m^{-4}) . \end{aligned} \quad (2.27)$$

with 9 coefficients c_a^n at α^n order.

Another constraint comes from the gauge symmetry which indicates the following Ward Identity (WI)

$$\mathcal{M}_v(k_1, k_2, q) = 0 \quad (2.28)$$

There are only 4 combinations of M_a that satisfy the WI, which are

$$\begin{aligned} \frac{1}{m}(p \cdot \epsilon)S_v &= M_1 + M_2 + \frac{v \cdot k}{m}M_1 = M_1(1 - \alpha_1 - \alpha_2) + M_2 , \\ \frac{1}{4m}T_v^{\rho\nu}(q_\nu\epsilon_\rho - q_\rho\epsilon_\nu) &= M_5 , \\ \frac{1}{4m^2}v^\rho k_\mu T_v^{\mu\nu}(q_\nu\epsilon_\rho - q_\rho\epsilon_\nu) &= M_3 - \frac{v \cdot q}{2m}M_6 = M_3 + M_6(\alpha_1 - \alpha_2) , \\ \frac{1}{4m^3}k_\perp^\rho k_\mu T_v^{\mu\nu}(q_\nu\epsilon_\rho - q_\rho\epsilon_\nu) &= M_4 - \frac{k_\perp \cdot q}{2m^2}M_6 = M_4 - M_6(\alpha_1 - \alpha_2) . \end{aligned} \quad (2.29)$$

The first makes use of the special kinematics $p \cdot q = \frac{1}{2}(m_1^2 - m_2^2) = 0$ for three point amplitude, while the rest are proportional to the field strength tensor $f_{\mu\nu} = q_\mu\epsilon_\nu - q_\nu\epsilon_\mu$ that guarantees gauge invariance. When WI is imposed on eq. (2.27), we find that

$$c_1 = c_2 , \quad c_1^1 = c_2^1 - 1 , \quad c_6^1 = c_3 - c_4 , \quad (2.30)$$

and only 6 independent parameters are retained.

$$\begin{aligned} \mathcal{M}_v(k_1, k_2, \epsilon) &= [c_1 + c_1^1(\alpha_1 + \alpha_2)](M_1 + M_2) + (\alpha_1 + \alpha_2)M_2 + [c_5 + c_5^1(\alpha_1 + \alpha_2)]M_5 \\ &\quad + c_3[M_3 + (\alpha_1 - \alpha_2)M_6] + c_4[M_4 - (\alpha_1 - \alpha_2)M_6] + O(m^{-4}) . \end{aligned} \quad (2.31)$$

If not for the RPI constraint, these are genuine independent parameters for the low energy effective theory, which do not necessarily have a Lorentz invariant UV completion.

Now let's see how the RPI condition eq. (2.20) works. First we can work out how the derivatives act on the fermion bilinears

$$\begin{aligned} \mathcal{D}_\perp^\rho S_v &\approx \frac{1}{4m^2} \left(2k_\perp^\rho S_v + T_v^{\rho\lambda} q_\lambda \right) , \\ \mathcal{D}_\perp^\rho T_v^{\mu\nu} &\approx \frac{1}{4m^2} \left(q_\perp^{[\mu} \eta_\perp^{\nu]\rho} S_v + 2k_\perp^\rho T_v^{\mu\nu} - 2\eta_\perp^{\rho[\mu} T_v^{\nu]\lambda} \mathbf{k}_{\perp\lambda} + 2T_v^{\rho[\mu} \mathbf{k}_\perp^{\nu]} + 4mT_v^{\rho[\mu} v^{\nu]} \right) . \end{aligned} \quad (2.32)$$

where higher order terms may be omitted. Besides, the reference velocity v^μ , the momentum k^μ and the order parameters $\alpha_{1,2}$ would vary under reparameterization

$$\begin{aligned} \mathcal{D}_\perp^\rho v^\lambda &= \frac{1}{m}(\delta^{\rho\lambda} - v^\rho v^\lambda) , \\ \mathcal{D}_\perp^\rho k_\perp^\lambda &= -(2 - \alpha_1 - \alpha_2)(\delta^{\rho\lambda} - v^\rho v^\lambda) - \frac{1}{m}k_\perp^\alpha v^\lambda , \\ \mathcal{D}_\perp^\rho \alpha_i &= -\frac{1}{2m^2}(1 - 2\alpha_i)k_{i\perp}^\alpha , \end{aligned} \quad (2.33)$$

while q^μ and ϵ^μ are invariant. It is easy to verify that the transverse derivative also satisfy the Leibniz rule $\mathcal{D}_\perp(fg) = (\mathcal{D}_\perp f)g + f(\mathcal{D}_\perp g)$. Hence we can solve eq.(2.20) for the generic three-point amplitude eq. (2.23),

$$\mathcal{M}_v = c_1 \mathcal{B}_1 + c_4 \mathcal{B}_2 . \quad (2.34)$$

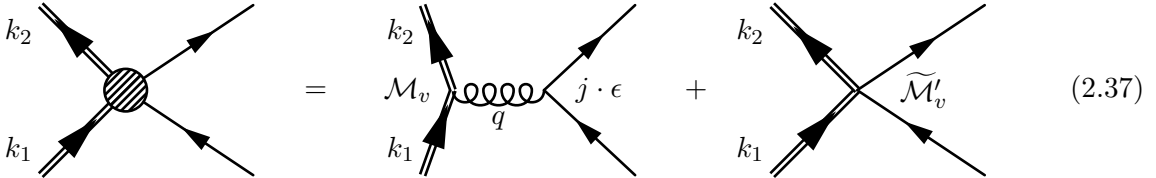
$$\begin{aligned} \mathcal{B}_1 &= M_1(1 - \frac{\alpha_1}{2} - \frac{\alpha_2}{2}) + M_2(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2}) - M_3 - M_5(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2}) - M_6(\alpha_1 - \alpha_2) , \\ \mathcal{B}_2 &= 2M_3 + M_4 + M_5(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2}) + M_6(\alpha_1 - \alpha_2) . \end{aligned} \quad (2.35)$$

Note that we don't need to assume $\mathcal{C}$ -even or gauge invariance here. The only two combinations that satisfy the RPI condition are automatically $\mathcal{C}$ -even and gauge invariant. It is a coincidence for the single gauge boson amplitude, as can be easily seen from the top down view that we show in the next section.

2.2.2 Amplitudes with a gauge current

During the evaluation of 3 point RPI amplitudes, we have been setting $q^2 = 0$. It is certainly true when we consider the amplitude with an on-shell gluon. However, the contributions proportional to q^2 will be important when we consider the amplitude between the heavy fermion and the current j_μ that minimally coupled to the gauge boson. The current j_μ include all kinds of particles charged under the gauge group. The minimal coupling is described by the on-shell amplitude $\mathcal{M}_0 = gj^\mu \epsilon_\mu$. Assuming that at low energy the heavy fermion and the current only interact through exchanging the gauge boson or some effective contact terms, then

$$\tilde{\mathcal{A}}_v(k_1, k_2, j) = \frac{1}{q^2} \tilde{\mathcal{M}}_v(k_1, k_2, j) + \tilde{\mathcal{M}}'_v(k_1, k_2, j) , \quad (2.36)$$



$$\text{Diagram (Left)} = \mathcal{M}_v \text{ (Diagram)} + \tilde{\mathcal{M}}'_v \text{ (Diagram)} \quad (2.37)$$

where both $\tilde{\mathcal{M}}$ and $\tilde{\mathcal{M}}'$ are local functions. Unitarity then demands

$$\begin{aligned} \text{Res}_{q^2=0} \tilde{\mathcal{A}} &= \tilde{\mathcal{M}}_v(k_1, k_2, j)|_{q^2=0} = \sum_s \mathcal{M}_v(k_1, k_2, \epsilon^{(s)}) \times (j^\mu \epsilon_\mu^{(-s)}) \\ &= \mathcal{M}_v(k_1, k_2, j) , \end{aligned} \quad (2.38)$$

where the spin sum is given by

$$\sum_s \epsilon_\mu^{(s)} \epsilon_\nu^{(-s)} = g_{\mu\nu} - q_\mu q_\nu / q^2 , \quad (2.39)$$

and the second term does not contribute due to the Ward Identity $\mathcal{M}_v(k_1, k_2, q) = 0$. The function $\mathcal{M}_v$ is the on-shell 3 point amplitude with gauge boson we derived previously, given

by the RPI combinations eq. (2.34). It shows that the new function $\widetilde{\mathcal{M}}_v$ should be the same as $\mathcal{M}_v$ when $q^2 = 0$, while in general they could differ by a term proportional to q^2 . But this term can always be absorbed into the definition of $\widetilde{\mathcal{M}}'_v$ in eq. (2.36). Hence we define $\widetilde{\mathcal{M}}_v = \mathcal{M}_v$ to have the same form, but with arbitrary q^2 . On the other hand, the local function $\widetilde{\mathcal{M}}'_v(k_1, k_2, j)$ should depend on the heavy spinor in the same form as eq. (2.21)

$$\widetilde{\mathcal{M}}'_v(k_1, k_2, j) = \tilde{A}(v, k, q, j)S_v + \tilde{B}_{\mu\nu}(v, k, q, j)T_v^{\mu\nu} , \quad (2.40)$$

where the current satisfies $q \cdot j = 0$ due to the conservation law, which plays the similar role as the on-shell condition for ϵ . Hence the coefficients $\tilde{A}, \tilde{B}$ have the same form factors as in eq. (2.22) with ϵ replaced by j , and C_a replaced by $\tilde{C}_a$ which can now depend on q^2 as well. In sum, we have a similar basis for it

$$\widetilde{\mathcal{M}}'_v(k_1, k_2, j) = \sum_{a=1}^6 \tilde{C}_a(\alpha_i) M_a(k_1, k_2, j) , \quad (2.41)$$

Now we can demand that $\tilde{\mathcal{A}}$ is RPI and thus satisfy $\mathcal{D}_\perp \tilde{\mathcal{A}} = 0$. It is crucial to notice that although $\widetilde{\mathcal{M}}_v$ is defined to be the same as $\mathcal{M}_v$ which is RPI, we don't necessarily have $\mathcal{D}_\perp \widetilde{\mathcal{M}}_v = 0$, because what we drop in $\mathcal{D}_\perp \mathcal{M}_v$ due to $q^2 = 0$ should be restored, and in general we could have $\mathcal{D}_\perp \widetilde{\mathcal{M}}_v \propto q^2$, which contributes a local term in $\mathcal{D}_\perp \tilde{\mathcal{A}}$

$$\mathcal{D}_\perp \tilde{\mathcal{A}} = \underbrace{\mathcal{D}_\perp \widetilde{\mathcal{M}}_v / q^2}_{\text{non-zero local term}} + \mathcal{D}_\perp \widetilde{\mathcal{M}}'_v = 0 \quad (2.42)$$

Up to $O(m^{-3})$, we solve the RPI condition and get

$$\widetilde{\mathcal{M}}'_v = \frac{c_1}{8m^2} M_1 - \frac{c_4}{4m^2} (M_1 + \frac{1}{2} M_2) + \frac{c'_2}{4m^2} (M_1 + M_2) + \frac{c'_5}{4m^2} M_5 . \quad (2.43)$$

All together, the RPI amplitude with a gauge current can be written as

$$\tilde{\mathcal{A}}(k_1, k_2, j) = c_1 \tilde{\mathcal{B}}_1 + c_4 \tilde{\mathcal{B}}_2 + c'_2 \tilde{\mathcal{B}}_3 + c'_5 \tilde{\mathcal{B}}_4 , \quad (2.44)$$

$$\begin{aligned} q^2 \tilde{\mathcal{B}}_1 &= M_1 \left(1 - \frac{\alpha_1}{2} - \frac{\alpha_2}{2} + \frac{q^2}{8m^2} \right) + M_2 \left(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2} \right) \\ &\quad - M_3 - M_5 \left(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2} \right) - M_6(\alpha_1 - \alpha_2) , \\ q^2 \tilde{\mathcal{B}}_2 &= -\frac{q^2}{8m^2} (2M_1 + M_2) + 2M_3 + M_4 + M_5 \left(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2} \right) + M_6(\alpha_1 - \alpha_2) , \\ \tilde{\mathcal{B}}_3 &= \frac{1}{4m^2} (M_1 + M_2) , \\ \tilde{\mathcal{B}}_4 &= \frac{1}{4m^2} M_5 , \end{aligned} \quad (2.45)$$

where M_a are given by eq. (2.24) with ϵ replaced by j .

One could easily recover the 3 point on-shell amplitudes $\mathcal{B}_{1,2}$ by taking the residues of $\tilde{\mathcal{B}}_a$ at $q^2 = 0$. $\tilde{\mathcal{B}}_{3,4}$ are new contact amplitudes that are proportional to q^2 , which do not have correspondence in the 3 point amplitude with gauge boson. We will see in the next section that they correspond to the redundant operators that can be removed by field redefinition.

2.3 Non-Linearly Realized Lorentz Symmetry

Since the RPI comes from Lorentz invariance in the UV, we can also construct the RPI amplitudes by starting from something explicitly Lorentz invariant using the variables $u(p_i)$ and the momentum $p_i = mv + k_i$. For example, the three point on-shell amplitudes have the following basis

$$\begin{aligned}\mathcal{B}_1 &= \epsilon_\mu \bar{u}(p_2) \gamma^\mu u(p_1) = \frac{1}{m} (p \cdot \epsilon) \bar{u}(p_2) u(p_1) - \mathcal{B}_2, \\ \mathcal{B}_2 &= \frac{1}{2m} q_\mu \epsilon_\nu \bar{u}(p_2) (-i\sigma^{\mu\nu}) u(p_1) .\end{aligned}\tag{2.46}$$

Reducing eq. (2.46) to the heavy quark spinors, we use eq. (2.7) with $f(k^2) = 1$

$$u(p_i) = \frac{1}{1 - k_i^2/4m^2} \left(1 + \frac{\not{k}_i}{2m} \right) u_v(k_i)\tag{2.47}$$

We can first compute the Lorentz covariant bilinears in terms of the heavy quark spinor bilinears $S_v, T_v^{\mu\nu}$:

$$\begin{aligned}\bar{u}(p_2) u(p_1) &= \frac{1}{(1 - k_2^2/4m^2)(1 - k_1^2/4m^2)} \bar{u}_v(k_2) \left(1 + \frac{\not{k}_2}{2m} \right) \left(1 + \frac{\not{k}_1}{2m} \right) u_v(k_1) \\ &= S_v \left(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2} - \frac{q^2}{8m^2} \right) + \frac{1}{4m^2} T_v^{\mu\nu} k_\mu q_\nu + O(m^{-3}), \\ \bar{u}(p_2) (-i\sigma^{\mu\nu}) u(p_1) &= \frac{1}{(1 - k_2^2/4m^2)(1 - k_1^2/4m^2)} \bar{u}_v(k_2) \left(1 + \frac{\not{k}_2}{2m} \right) (-i\sigma^{\mu\nu}) \left(1 + \frac{\not{k}_1}{2m} \right) u_v(k_1) \\ &= T_v^{\mu\nu} \left(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2} - \frac{q^2}{8m^2} \right) + \frac{1}{m} v^{[\mu} T_v^{\nu]\rho} k_\rho - \frac{1}{2m} q^{[\mu} v^{\nu]} S_v \\ &\quad + \frac{1}{4m^2} \left[k^{[\mu} q^{\nu]} S_v + 2k^{[\mu} T_v^{\nu]\rho} k_\rho - \frac{1}{2} q^{[\mu} T_v^{\nu]\rho} q_\rho \right] + O(m^{-3})\end{aligned}\tag{2.48}$$

where we used the Schouten identity for $D = 4$ and the definition $T_v^{\mu\nu} = i\epsilon^{\mu\nu\alpha\beta} v_\alpha \bar{u}_v \gamma_\beta \gamma^5 u_v$. We also have

$$\begin{aligned}\frac{1}{m} p \cdot \epsilon &= v \cdot \epsilon + \frac{1}{m} k_\perp \cdot \epsilon + \frac{k \cdot v}{m} (v \cdot \epsilon) \\ &= (v \cdot \epsilon) (1 - \alpha_1 - \alpha_2) + \frac{1}{m} k_\perp \cdot \epsilon + O(m^{-4}).\end{aligned}\tag{2.49}$$

Plugging them in eq. (2.46) and taking $q^2 = 0$, we get exactly the same amplitudes as eq. (2.34) at the leading orders.

Note that we also deliberately retain the q^2 dependency in eq. (2.48), since the bilinears $S_v, T_v^{\mu\nu}$ are defined for arbitrary q^2 . When we compute the tree level amplitude $\tilde{\mathcal{A}}(k_1, k_2, j)$ in eq. (2.36), they will contribute to the first term and result in $\tilde{\mathcal{B}}_{1,2}$ in eq. (2.45). The

additional Lorentz invariant local amplitudes between the fermion and a gauge current have the following complete basis

$$\tilde{\mathcal{B}}_3 = \frac{1}{4m^3}(p \cdot j)\bar{u}(p_2)u(p_1) , \quad \tilde{\mathcal{B}}_4 = \frac{1}{8m^3}q_\mu j_\nu \bar{u}(p_2)(-i\sigma^{\mu\nu})u(p_1) . \quad (2.50)$$

Substitute in the low energy variables, to leading order we get precisely the $\tilde{\mathcal{B}}_{3,4}$ in eq. (2.45).

The advantage of the top-down approach is that we do not need to solve the RPI condition eq. (2.20). As long as we can get the complete basis of the Lorentz invariant amplitudes, the heavy particle amplitudes in the low energy theory would be obtained directly. The 3 point on-shell amplitudes have been completely studied in [25] by using the massive spinor variables

$$u(p_i) = \begin{pmatrix} |\mathbf{i}\rangle \\ |\mathbf{i}| \end{pmatrix} , \quad \bar{u}(p_i) = \left(\langle \mathbf{i}| \quad [\mathbf{i}] \right) , \quad (2.51)$$

where the bolded helicity spinors are doublets under the little group $SU(2)$. For the amplitudes with an on-shell gauge boson, a complete basis is given by

$$x\langle \mathbf{21} \rangle = (p \cdot \epsilon)\bar{u}(p_2)u(p_1) , \quad \frac{1}{2m}\langle \mathbf{23} \rangle \langle \mathbf{31} \rangle = \frac{1}{2m}q_\mu \epsilon_\nu \bar{u}(p_2)(-i\sigma^{\mu\nu})u(p_1) , \quad (2.52)$$

which are exactly the on-shell basis we started with. For on-shell amplitudes with more than 3 points, a complete basis can be obtained by the Young Tensor method up to any mass dimensions [1, 15]. However, the current j_μ is not an on-shell particle; we can use a current of an auxilliary particle to construct the basis. For example, we use an on-shell fermion current $j_\mu = \langle 3|\sigma_\mu|4\rangle$ with massless fermions $\psi_3, \psi_4^\dagger$ with opposite helicities (we use ψ to represent $h = -1/2$ spinors, and thus $\psi^\dagger$ has $h = +1/2$). We should be examining the 4 point amplitude $\mathcal{A}(k_1, k_2, \psi_3, \psi_4^\dagger)$. The enumeration of the massive amplitude basis starts from listing all the combinations of helicities of the massive particles $u(p_1), u(p_2)$. Up to the desired order, we have

$$\begin{aligned} (-, -) : & \langle \mathbf{21} \rangle \langle 3|p_2|4\rangle , & (+, +) : & [\mathbf{21}] \langle 3|p_2|4\rangle , \\ (-, +) : & \langle \mathbf{13} \rangle [\mathbf{24}] , & (+, -) : & [\mathbf{14}] \langle \mathbf{23} \rangle . \end{aligned} \quad (2.53)$$

Note that for each combination, we are using the Young tensor method to give a complete basis at given order, which generically only applies to massless amplitudes. For example, the $(-, -)$ type could have two different local amplitudes

$$\tilde{\mathcal{B}}_1^{--} = \langle \mathbf{21} \rangle \langle 3|p_2|4\rangle , \quad \tilde{\mathcal{B}}_2^{--} = \langle \mathbf{23} \rangle \langle \mathbf{31} \rangle [34] . \quad (2.54)$$

Note that their massless versions are equivalent

$$\langle 21 \rangle \langle 3|p_2|4\rangle = \langle 23 \rangle \langle 1|p_2|4\rangle + \langle 31 \rangle \langle 2|p_2|4\rangle = -\langle 23 \rangle \langle 1|p_3|4\rangle = \langle 23 \rangle \langle \mathbf{31} \rangle [34] \quad (2.55)$$

where we use the Schouten identity $\langle ij \rangle \langle k| = \langle ik \rangle \langle j| + \langle kj \rangle \langle i|$ for the first equation, momentum conservation $p_2 = -p_1 - p_3 - p_4$ and on-shell condition $\langle 11 \rangle = \langle 22 \rangle = [44] = 0$ for the second

condition. The Young tensor method directly provides the semi-standard Young tableau $\begin{array}{|c|c|c|} \hline \bar{1} & \bar{1} & 2 \\ \hline 2 & 3 & 3 \\ \hline \end{array} \simeq [34]\langle 13\rangle\langle 23\rangle$ as the only independent amplitude⁴, coincide with the equivalence proved above. For the massive version, the on-shell condition should be modified by the Dirac equation

$$p_{i\mu}\sigma^\mu|\mathbf{i}\rangle = m|\mathbf{i}\rangle, \quad (2.56)$$

thus we have

$$\begin{aligned} \tilde{\mathcal{B}}_1^{--} &= \langle 23\rangle\langle 1|p_2|4\rangle + \langle 31\rangle\langle 2|p_2|4\rangle \\ &= -\langle 23\rangle\langle 1|(p_1+p_3)|4\rangle + \langle 31\rangle\langle 2|p_2|4\rangle \\ &= \tilde{\mathcal{B}}_2^{--} - m([14]\langle 23\rangle + \langle 13\rangle[24]). \end{aligned} \quad (2.57)$$

Their difference can be found in the $(+, -)$ and $(-, +)$ type. In general, the difference between massive amplitudes that are equivalent in the massless limit must belong to amplitudes at lower mass dimensions, probably of other helicity types. One of the core idea of on-shell amplitude basis and Young tensor method is that we count the independent basis type by type, so if we had constructed the complete basis of the lower dimensional types, we should not consider $\tilde{\mathcal{B}}_{1,2}^{--}$ as independent amplitudes any more. Following this logic, we can still use the Young tensor method to construct the massive amplitude basis as in eq. (2.53)⁵. These four massive amplitudes can be combined into the parity even/odd ones

$$\begin{aligned} \tilde{\mathcal{B}}_3^{\text{even/odd}} &= \frac{1}{4m^3}([\mathbf{21}] \pm \langle \mathbf{21}\rangle)\langle 3|p_2|4\rangle = \frac{1}{4m^3}(p \cdot j)\bar{u}(p_2)(\gamma^5)u(p_1), \\ \tilde{\mathcal{B}}_4^{\text{even/odd}} &= \frac{1}{8m^3}([24][41]\langle 34\rangle \pm \langle 23\rangle\langle 31\rangle[34]) = \frac{1}{8m^3}q_\mu j_\nu \bar{u}(p_2)(-i\sigma^{\mu\nu})(\gamma^5)u(p_1). \end{aligned} \quad (2.58)$$

The parity even ones are exactly eq. (2.50), and hence reduce to the last two in eq. (2.45).

One could go to higher dimensions with Young tensor method, but to restrict to the amplitude with current $j_\mu = \langle 3|\sigma_\mu|4\rangle$ instead of arbitrary amplitudes with the two fermions $\psi_3, \psi_4^\dagger$, it is necessary to keep track of the total angular momentum $J_{\{3,4\}}$ of the two particles. Only the $J_{\{3,4\}} = 1$ amplitudes are relevant to processes in eq. (2.37) that we are interested. To pick out these amplitudes, we need to apply the $\mathbf{W}^2$ Casimir operator on the amplitude basis to find the partial waves, as explained in [27, 28].

We can also consider amplitudes with two on-shell gauge bosons, which also start at dimension 7. We need to consider the cases of same-helicity and opposite-helicity gluons. For

⁴According to the amplitude basis of the helicity type $(-1/2, -1/2, -1/2, +1/2)$ is given by semi-standard Young tableau with numbers $\{1, 1, 2, 2, 3, 3\}$ filling in the Young diagram $\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline & & \\ \hline \end{array}$ with $\bar{n} = 1$ column for square bracket and $n = 2$ columns for angle brackets. We are using the numbers with bar to denote the square bracket columns. Hence the column $\begin{array}{|c|} \hline \bar{1} \\ \hline \end{array}$ represents $\frac{1}{2}\epsilon^{12ij}[ij] = [34]$, while $\begin{array}{|c|c|} \hline \bar{1} & \bar{2} \\ \hline \end{array}$ represent $\langle 13\rangle\langle 23\rangle$.

⁵The Young tensor method is applicable for massive fermions [10]. For higher spin massive particles, the massless limit of intermediate helicities is trickier. For example, the massless limit of the longitudinal polarized vector particle is similar to the amplitudes with a Goldstone boson with Adler zero condition [7, 8]. We are not generalizing the top-down method to massive particles with arbitrarily spins.

the same-helicity gluons $\mathcal{M}(\psi_1, \psi_2, g_3^+, g_4^+)$, we have the amplitude basis

$$\mathcal{B}_5^{\text{even/odd}} = \frac{1}{m^3}([\mathbf{12}] \pm \langle \mathbf{12} \rangle)[34]^2, \quad \mathcal{B}_6 = \frac{1}{m^3}([\mathbf{13}][\mathbf{24}] + [\mathbf{14}][\mathbf{23}])[34]. \quad (2.59)$$

Since the spin-statistics requires the symmetry between the two gauge bosons, we are forced to choose the symmetric/anti-symmetric color structures for $\mathcal{B}_{5,6}$. For $U(1)$ photons, only $\mathcal{B}_5$ survives. For the color $SU(3)$, we may use $d^{abc}T^c$ or δ^{ab} for $\mathcal{B}_5$, and f^{abc} for $\mathcal{B}_6$. In addition, we have the following basis of opposite-helicity amplitudes $\mathcal{M}(\psi_1, \psi_2, g_3^-, g_4^+)$ at leading order:

$$\begin{aligned} \mathcal{B}_7^{\text{even/odd}} &= \frac{1}{m^4}(\langle \mathbf{13} \rangle[\mathbf{24}] \pm \langle \mathbf{23} \rangle[\mathbf{14}])\langle 3|p_2|4 \rangle, \\ \mathcal{B}_8^{\text{even/odd}} &= \frac{1}{m^5}([\mathbf{21}] \pm \langle \mathbf{12} \rangle)\langle 3|p_2|4 \rangle^2. \end{aligned} \quad (2.60)$$

The reason that we go to the $O(m^{-5})$ order is that after taking the non-relativistic limit, the momenta $p_2 = mv + k_2$ would provide a factor of m at leading order, and hence they both contribute at $O(m^{-3})$ order of the HQET. One may check that the amplitudes $\mathcal{B}_{5,6,7,8}$ correspond to the $g^2 m^{-3}$ order Lagrangian terms in the HQET as shown in [18].

3 Matching with the Lagrangian

The RPI local amplitudes indeed correspond to a RPI effective Lagrangian of the HPET. The easiest way to show it is to find the corresponding operators of the Lorentz invariant amplitudes in eq. (2.46) and (2.50)

$$\begin{aligned} \mathcal{Q}_1 &= \bar{\psi} i \not{D} \psi, \\ \mathcal{Q}_2 &= \frac{1}{4m} G_{\mu\nu} \bar{\psi} \sigma^{\mu\nu} \psi, \\ \tilde{\mathcal{Q}}_3 &= \frac{1}{8m^3} \bar{\psi} (i \overleftrightarrow{D}_\mu) \psi J^\mu, \\ \tilde{\mathcal{Q}}_4 &= \frac{1}{8m^3} \bar{\psi} \sigma^{\mu\nu} \psi D_\mu J_\nu. \end{aligned} \quad (3.1)$$

where the current operator can be converted to the gauge field by EOM

$$J_\mu \simeq D^\nu G_{\mu\nu}, \quad (3.2)$$

and the covariant derivative is defined as

$$D_\mu = \partial_\mu + igZ A_\mu \quad (3.3)$$

so that we have

$$\begin{aligned} \mathcal{Q}_3 &= -\frac{ig}{8m^3} \bar{\psi} \{iD_\mu, [iD_\nu G^{\mu\nu}]\} \psi, \\ \mathcal{Q}_4 &= -\frac{g}{8m^3} \bar{\psi} \sigma^{\mu\nu} [iD_\mu, [iD^\rho, G_{\nu\rho}]] \psi. \end{aligned} \quad (3.4)$$

The field $\psi(x)$ transforms covariantly under the Lorentz group, and correspond to the wave function $u(p)$ in the amplitude. Hence these operators directly produce the RPI amplitudes $\mathcal{B}_{1,2,3,4}$ we derived in the last section. However, they are not yet written in terms of the low energy heavy fermion fields $\psi_v(x)$, and not valid as a perturbation expansion for large m . The heavy fermion field $\psi_v(x)$ should have the defining feature

$$\not{p}\psi_v(x) = \psi_v(x) , \quad (3.5)$$

and consequently they produce the wave function $u_v(k)$ in the amplitude. One trick adopted in Ref. [29] is to define Reparametrization Invariant Spinor Field $\Psi_v \equiv \Gamma(v, iD)\psi_v$ such that under a reparameterization $v \rightarrow w = v + \ell/m$

$$\Psi_w = e^{i\ell \cdot x} \Psi_v . \quad (3.6)$$

The Invariant Operator $\Gamma(v, iD)$ has to satisfy the “invariance equation”

$$\Gamma(w, iD - \ell) \mathcal{B}^{-1} \tilde{W}(\mathcal{B}, iD + mv) = \Gamma(v, iD) \quad (3.7)$$

where $v^\mu = \mathcal{B}^\mu_\nu w^\nu$ and $\tilde{W}$ is the $\mathcal{B}$ induced little group transformation with respect to the physical momentum $p = mv + k$. Note that $\tilde{W}$ depends on the definition of the field $\psi_v(x)$, or more specifically on its normalization, up to which it is defined in eq. (3.5). It is the same ambiguity of normalization $f(k^2)$ we discussed for the wave function $u_v(k)$. In [29], $\tilde{W} \equiv N^{-1} W N$ where $N(v, iD)$ is exactly a conventional non-relativistic normalization of the heavy field. We are leaving this normalization unspecified for now.

Eq.(3.6) indicates that $e^{-imv \cdot x} \Psi_v$ is RPI, and should generate the heavy fermion with the RPI wave function $u(p)$, making it a low energy version of the field $\psi(x)$ (but with the anti-particle integrated out). In that $iD_\mu e^{-imv \cdot x} \Psi_v = e^{-imv \cdot x} (mv + iD)_\mu \Psi_v$, the RPI operators can be written as

$$\begin{aligned} \mathcal{Q}_1 &= m \bar{\Psi}_v (\not{V} - 1) \Psi_v , \\ \mathcal{Q}_2 &= \frac{g}{4m} \bar{\Psi}_v \sigma^{\mu\nu} G_{\mu\nu} \Psi_v , \\ \mathcal{Q}_3 &= -\frac{ig}{16m^2} \bar{\Psi}_v \{ \mathcal{V}_\mu, [m \mathcal{V}_\nu, G^{\mu\nu}] \} \Psi_v , \\ \mathcal{Q}_4 &= -\frac{g}{8m^3} \bar{\Psi}_v [m \mathcal{V}_\mu, \sigma^{\mu\nu} [m \mathcal{V}^\rho, G_{\nu\rho}]] \Psi_v . \end{aligned} \quad (3.8)$$

with $\mathcal{V}_\mu \equiv v_\mu + iD_\mu/m$. After solving eq. (3.7) for the function Γ , we can substitute $\Psi_v = \Gamma(v, iD)\psi_v$ and obtain 4 combinations of operators in terms of ψ_v . Their contribution to the heavy fermion amplitudes, on the other hand, would be precisely how the Lorentz invariant amplitudes $\mathcal{B}_{1,2,3,4}$ in eq. (2.46) and (2.50) reduce to the heavy fermion amplitudes.

We proved the RPI of eq. (2.34) and (2.45) which used the normalization $f(k^2) = 1$. But as we mentioned earlier, different field redefinitions N and Γ lead to different normalization

$f(k^2)$ of the wave function u_v . We present two important examples here, starting from the Lagrangian

$$\mathcal{L} = \mathcal{Q}_1 + a_F \mathcal{Q}_2 + a_D \mathcal{Q}_3 + a_{W_1} \mathcal{Q}_4 \quad (3.9)$$

For example, a field redefinition based on $N = \left(\frac{m^2}{m^2 + D_\perp^2} \right)^{1/4}$ leads to [29]⁶

$$\begin{aligned} \Gamma^{(1)} = & 1 + \frac{i \not{D}_\perp}{2m} + \frac{(i D_\perp)^2}{8m^2} - g \frac{\sigma \cdot G}{16m^2} - \frac{i \not{D}_\perp i v \cdot D}{2m^2} + g \frac{v^\mu v_\rho (-i \sigma^{\rho\nu}) i G_{\mu\nu}}{8m^2} \\ & - \frac{i \not{D}_\perp (i D_\perp)^2}{16m^3} + \frac{i \not{D}_\perp (i v \cdot D)^2}{2m^3} + g \frac{\sigma \cdot G i \not{D}_\perp}{16m^3} - g \frac{i \not{D}_\perp \sigma \cdot G}{32m^3} \\ & - g \frac{v^\mu v_\rho (-i \sigma^{\rho\nu}) i G_{\mu\nu} i \not{D}_\perp}{8m^3} + g \frac{v^\mu v_\rho i \not{D}_\perp (-i \sigma^{\rho\nu}) i G_{\mu\nu}}{16m^3} \\ & + (a_F + 1) g \frac{v^\mu i D_\perp^\nu i G_{\mu\nu}}{4m^3} + (a_F + 1) g \frac{v^\rho (-i \sigma^{\mu\nu}) i D_{\perp\mu} i G_{\rho\nu}}{4m^3} \\ & + a_D g \frac{i G_{\mu\nu} v^\mu i D_\perp^\nu}{8m^3} + a_{W_1} g \frac{v^\rho (-i \sigma^{\mu\nu}) [i D_{\perp\mu}, i G_{\rho\nu}]}{8m^3} \end{aligned} \quad (3.10)$$

while the operators are redefined in terms of $\psi_v^{(1)}$, such as

$$\begin{aligned} \mathcal{Q}_1 &= \bar{\psi}_v^{(1)} \Gamma^{(1)}(v, -iD) (i \not{D}) \Gamma^{(1)}(v, iD) \psi_v^{(1)} \\ &= \bar{\psi}_v \left\{ i v \cdot D + \frac{(i D_\perp)^2}{2m} + \frac{(i D_\perp)^4}{8m^3} - g \frac{\sigma \cdot G}{4m} + \dots \right\} \psi_v. \end{aligned} \quad (3.11)$$

This particular redefinition leads, up to the order $O(m^{-3})$, to the canonical form of HQET

$$\begin{aligned} \mathcal{L} = & \bar{\psi}_v^{(1)} \left\{ i v \cdot D - c_2 \frac{D_\perp^2}{2m} + c_4 \frac{D_\perp^4}{8m^3} - c_F g \frac{\sigma_{\mu\nu} G^{\mu\nu}}{4m} - c_D g \frac{v_\mu (D_{\perp\nu} G^{\mu\nu})}{8m^2} + i c_S g \frac{v_\mu \sigma_{\nu\rho} \{D_\perp^\nu, G^{\mu\rho}\}}{8m^2} \right. \\ & + c_{W_1} g \frac{\{D_\perp^2, \sigma_{\mu\nu} G^{\mu\nu}\}}{16m^3} - c_{W_2} g \frac{D_{\perp\rho} \sigma_{\mu\nu} G^{\mu\nu} D_\perp^\rho}{8m^3} + c_{p'p} g \frac{\sigma_{\mu\nu} (D_{\perp\rho} G^{\rho\mu} D_\perp^\nu + D_\perp^\nu G^{\rho\mu} D_{\perp\rho} - D_{\perp\rho} G^{\mu\nu} D_\perp^\rho)}{8m^3} \\ & \left. - i c_M g \frac{D_{\perp\mu} (D_{\perp\nu} G^{\mu\nu}) + (D_{\perp\nu} G^{\mu\nu}) D_{\perp\mu}}{8m^3} \right\} \psi_v^{(1)} \end{aligned} \quad (3.12)$$

where $c_2 = c_4 = 1$, $c_F = a_F$, $c_D = a_D$, $c_S = 2a_F - 1$, $c_{W_1} = a_{W_1}$, $c_{W_2} = a_{W_1} - 1$, $c_{p'p} = a_F - 1$, $c_M = (a_D - a_F)/2$, which are precisely the well-known constraints from RPI. From this Lagrangian, one can easily derive the contributions to the heavy fermion amplitude with one gluon in terms of $u_v^{(1)}(k)$, with the normalization $f^{(1)}(k^2) = 1 - k^2/8m^2$, as

$$\mathcal{B}_1^{(1)} = M_1 + M_2(1 + \alpha_1 + \alpha_2) - M_3 - M_5(1 + \alpha_1 + \alpha_2) - M_6(\alpha_1 - \alpha_2) + O(\alpha^2). \quad (3.13)$$

where the building blocks M_i are all defined in terms of $u_v^{(1)}(k)$.

⁶We used $Z = 1$ here to make the comparison with [18] more clear.

$$\begin{aligned}
\mathcal{Q}_1 = \bar{\psi}_v \Bigg\{ & iv \cdot D + \frac{(iD_\perp)^2}{2m} + \frac{(iD_\perp)^4}{8m^3} - g \frac{\sigma \cdot G}{4m} + g \frac{v^\mu [iD_\perp^\nu, iG_{\mu\nu}]}{8m^2} \\
& - g(-i\sigma^{\mu\nu})v^\rho \frac{\{iD_{\perp\mu}, iG_{\nu\rho}\}}{8m^2} - g \frac{\{iD_\perp^2, \sigma \cdot G\}}{16m^3} \\
& + a_F g \frac{iv \cdot D v^\mu iD_\perp^\nu iG_{\mu\nu} - iG_{\mu\nu} v^\mu iD_\perp^\nu iv \cdot D}{4m^3} \\
& + a_F g(-i\sigma^{\mu\nu})v^\rho \frac{iv \cdot D iD_{\perp\mu} iG_{\rho\nu} + iG_{\rho\nu} iD_{\perp\mu} iv \cdot D}{4m^3} \\
& + a_D g \frac{iv \cdot D iG_{\mu\nu} v^\mu iD_\perp^\nu - v^\mu iD_\perp^\nu iG_{\mu\nu} iv \cdot D}{8m^3} \\
& + a_{W_1} g(-i\sigma^{\mu\nu})v^\rho \frac{[iv \cdot D, [iD_{\perp\mu}, iG_{\rho\nu}]]}{8m^3} \Bigg\} \psi_v + \mathcal{O}(g^2).
\end{aligned} \tag{3.14}$$

In the language of [18], the canonical form of HQET has

$$\begin{aligned}
\mathcal{Q}_1^{(1)} = \bar{\psi}_v \Bigg\{ & iv \cdot D + \mathcal{O}_2 + \mathcal{O}_4 + \mathcal{O}_F + \mathcal{O}_D + \mathcal{O}_S + \mathcal{O}_{W_1} \\
& + a_F g \left[v^\mu \frac{iv \cdot D iD_\perp^\nu iG_{\mu\nu} - iG_{\mu\nu} iD_\perp^\nu iv \cdot D}{4m^3} \right. \\
& + (-i\sigma^{\mu\nu})v^\rho \frac{iv \cdot D iD_{\perp\mu} iG_{\rho\nu} + iG_{\rho\nu} iD_{\perp\mu} iv \cdot D}{4m^3} \Bigg] \\
& + a_D g v^\mu \frac{iv \cdot D iG_{\mu\nu} iD_\perp^\nu - iD_\perp^\nu iG_{\mu\nu} iv \cdot D}{8m^3} \\
& + a_{W_1} g(-i\sigma^{\mu\nu})v^\rho \frac{[iv \cdot D, [iD_{\perp\mu}, iG_{\rho\nu}]]}{8m^3} \Bigg\} \psi_v + \mathcal{O}(g^2).
\end{aligned} \tag{3.15}$$

which contributes to the heavy fermion amplitude in terms of $u_v(k)$, with normalization $f^{(1)}(k^2) = 1 - \frac{k^2}{8m^2}$, as

$$\mathcal{B}_1^{(1)} = M_1 + M_2(1 + \alpha_1 + \alpha_2) - M_3 - M_5(1 + \alpha_1 + \alpha_2) - M_6(\alpha_1 - \alpha_2) + \mathcal{O}(\alpha^2). \tag{3.16}$$

Similarly, we can derive the other RPI operators based on eq. (3.10) as

$$\begin{aligned}
\mathcal{Q}_2^{(1)} = g\bar{\psi}_v \Bigg\{ & -\frac{\sigma \cdot G}{4m} + \frac{v^\mu [iD_\perp^\nu, iG_{\mu\nu}]}{4m^2} - (-i\sigma^{\mu\nu})v^\rho \frac{\{iD_{\perp\mu}, iG_{\nu\rho}\}}{4m^2} \\
& - \frac{\{(iD_\perp)^2, \sigma \cdot G\}}{32m^3} - (-i\sigma^{\nu\rho}) \frac{iD_\perp^\mu iG_{\mu\nu} iD_{\perp\rho} + iD_{\perp\rho} iG_{\mu\nu} iD_\perp^\mu}{8m^3} \\
& - \frac{iD_{\perp\mu} \sigma \cdot G iD_\perp^\mu}{16m^3} + \frac{\{iD_\perp^\mu, [iD_\perp^\nu, iG_{\mu\nu}]\}}{16m^3} \\
& - \frac{iv \cdot D v^\mu iD_\perp^\nu iG_{\mu\nu} - iG_{\mu\nu} v^\mu iD_\perp^\nu iv \cdot D}{4m^3} \\
& - (-i\sigma^{\mu\nu})v^\rho \frac{iv \cdot D iD_{\perp\mu} iG_{\rho\nu} + iG_{\rho\nu} iD_{\perp\mu} iv \cdot D}{4m^3} \Bigg\} \psi_v + \mathcal{O}(g^2)
\end{aligned} \tag{3.17}$$

$$\begin{aligned}
\mathcal{Q}_3^{(1)} &= g\bar{\psi}_v \left\{ \frac{v^\mu [iD_\perp^\nu, iG_{\mu\nu}]}{8m^2} + \frac{iD_{\perp\mu} iG^{\mu\nu} iD_{\perp\nu}}{8m^3} - \frac{iv \cdot D iG_{\mu\nu} v^\mu iD_\perp^\nu - v^\mu iD_\perp^\nu iG_{\mu\nu} iv \cdot D}{8m^3} \right\} \psi_v + \mathcal{O}(g^2) \\
&= g\bar{\psi}_v \left\{ \mathcal{O}_D - \frac{1}{2} \mathcal{O}_M - \frac{iv \cdot D iG_{\mu\nu} v^\mu iD_\perp^\nu - v^\mu iD_\perp^\nu iG_{\mu\nu} iv \cdot D}{8m^3} \right\} \psi_v + \mathcal{O}(g^2)
\end{aligned} \tag{3.18}$$

$$\mathcal{Q}_4^{(1)} = g\bar{\psi}_v \left\{ \frac{[iD_{\perp\mu}, [iD_\perp^\mu, \sigma \cdot G]]}{16m^3} - (-i\sigma^{\mu\nu})v^\rho \frac{[iv \cdot D, [iD_{\perp\mu}, iG_{\rho\nu}]]}{8m^3} \right\} \psi_v + \mathcal{O}(g^2) \tag{3.19}$$

We can make other choices of field redefinitions. For instance, based on the RPI spinor fields defined in [18]⁷ where $u_v(k)$ is a k -dependent Lorentz transformation of $u(p)$. As derived in eq. (2.10), we have

$$f^{(2)}(k^2) = \left(1 - \frac{k^2}{4m^2}\right)^{-1/2}. \tag{3.20}$$

The corresponding field redefinition can be solved as

$$\Gamma^{(2)} = 1 + \frac{i\not{D}_\perp}{2m} - \frac{(iD_\perp)^2}{8m^2} - \frac{i\not{D}_\perp iv \cdot D}{2m^2} + \frac{(iD_\perp)^2 iv \cdot D}{4m^3} - 3 \frac{i\not{D}_\perp (iD_\perp)^2}{16m^3} + \frac{i\not{D}_\perp (iv \cdot D)^2}{2m^3} \tag{3.21}$$

which maps the same operator $\mathcal{O}_1$ to

$$\begin{aligned}
\mathcal{Q}_1^{(2)} = \bar{\psi}_v \Bigg\{ & iv \cdot D + \frac{(iD_\perp)^2}{2m} - \frac{(iD_\perp)^4}{8m^3} + g \frac{iG_{\mu\nu} (-i\sigma^{\mu\nu})}{4m} - \frac{\{iv \cdot D, (iD_\perp)^2\}}{8m^2} - \frac{iD_{\perp\mu} iv \cdot D iD_\perp^\mu}{4m^2} \\
& + g \frac{\{iD_{\perp\mu}, iG_{\nu\rho}\} (-i\sigma^{\mu\nu}) v^\rho}{8m^2} - g \frac{\{iv \cdot D, iG_{\mu\nu}\} (-i\sigma^{\mu\nu})}{16m^2} - g \frac{\{(iD_\perp)^2, iG_{\mu\nu}\} (-i\sigma^{\mu\nu})}{32m^3} \Bigg\} \psi_v
\end{aligned} \tag{3.22}$$

⁷Eq. (65) should be $\Psi_v = \left[A + \frac{i\not{D}_\perp}{2m} B\right] \psi_v$

	$f(k^2)$	$\mathcal{O}_1$
Canonical [18]	$1 - \frac{k^2}{2m^2}$	$g\{-Z[M_1 + M_2(1 + \alpha_1 + \alpha_2)] + M_3 + M_5(1 + \alpha_1 + \alpha_2) + M_6(\alpha_1 - \alpha_2)\}$
RPI Spinor [18]	$1 + \frac{k^2}{2m^2}$	$-gZ[M_1(1 - \alpha_1 - \alpha_2) + M_2 - M_3 - M_5 + M_6(\alpha_1 - \alpha_2)]$
Bottom-Up	1	$-gZ[M_1(1 - \frac{\alpha_1}{2} - \frac{\alpha_2}{2}) + M_2(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2}) - M_3 - M_5(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2}) + M_6(\alpha_1 - \alpha_2)]$

Table 1. aaa[18]

The resulting heavy fermion amplitude is thus

$$\mathcal{B}_1^{(2)} = M_1(1 - \alpha_1 - \alpha_2) + M_2 - M_3 - M_5 - M_6(\alpha_1 - \alpha_2) + O(\alpha^2), \quad (3.23)$$

which differs from eq. (3.16) only by a factor of $(1 - \alpha_1 - \alpha_2)$ up to higher orders. The factor exactly matches with the change of normalization

$$\mathcal{B}_i = \mathcal{B}_i^{(1)} f^{(1)}(k_1^2) f^{(1)}(k_2^2) \simeq \mathcal{B}_i^{(2)} f^{(2)}(k_1^2) f^{(2)}(k_2^2) \quad (3.24)$$

Following is the amplitudes in different normalization scenarios:

	$f(k^2)$	$\mathcal{O}_2$
Canonical	$1 - \frac{k^2}{2m^2}$	$2M_3 + M_4 + M_5(1 + \alpha_1 + \alpha_2) + M_6(\alpha_1 - \alpha_2)$
RPI Spinor	$1 + \frac{k^2}{2m^2}$	$-2M_3 - M_4 - M_5 - M_6(\alpha_1 - \alpha_2)$
Bottom-Up	1	$-2M_3 - M_4 - M_5(1 + \frac{\alpha_1}{2} + \frac{\alpha_2}{2}) - M_6(\alpha_1 - \alpha_2)$

4 Conclusion

In this work, we have systematically investigated the reparametrization invariance (RPI) of the Heavy Quark Effective Theory (HQET) from the perspective of scattering amplitudes. Our approach provides a novel framework for constructing and analyzing heavy quark amplitudes while ensuring consistency with the symmetries and constraints of the effective field theory.

We developed a bottom-up method to construct heavy quark amplitudes involving gluons, both on-shell and off-shell, using the building blocks of heavy spinor variables S_v and $T_v^{\mu\nu}$. These amplitudes were organized into a basis that respects the on-shell conditions and power counting in $1/m_Q$. By imposing gauge invariance and charge conjugation symmetry, we identified a minimal set of independent amplitudes up to $O(m_Q^{-3})$. The RPI condition, which ensures that physical observables are independent of the choice of the reference velocity v , was rigorously applied to the amplitude basis. We derived the constraints imposed by RPI and showed how they reduce the number of independent parameters in the low-energy effective theory. Notably, the RPI condition automatically enforces gauge invariance and selects specific combinations of amplitudes that correspond to the known HQET operators. We established a direct mapping between the RPI amplitudes and the effective operators in the HQET Lagrangian. By comparing our amplitude-based results with the canonical HQET

Lagrangian, we demonstrated consistency and highlighted the role of field redefinitions and normalization choices. For instance, different normalizations of the heavy spinor wave functions, lead to equivalent physical predictions but differ in their intermediate expressions. To further validate our results, we adopted a top-down approach, starting from Lorentz-invariant amplitudes in the full theory and reducing them to the heavy quark effective theory. This approach confirmed the RPI combinations derived from the bottom-up construction and provided additional insights into the structure of higher-dimensional operators.

Our work bridges the gap between on-shell amplitude methods and effective field theories, offering a powerful tool for studying heavy quark systems and other non-relativistic effective theories. The techniques developed here can be extended to other scenarios, such as dark matter effective theories or black hole physics. Future research could also explore the generalization of the amplitude basis to higher orders in $1/m_Q$, and to include interactions with multiple gluons or other light degrees of freedom, enabling a more comprehensive study of heavy particle dynamics. In summary, this work provides a robust and systematic framework for studying HQET and other non-relativistic effective theories using on-shell amplitudes. By leveraging reparametrization invariance, we have derived a consistent and computationally efficient description of the heavy particle interactions, paving the way for further advancements in both theoretical and phenomenological studies.

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